

Paying the Price

Rahul Lal, Rayaam Damani, Kyle Leung



Presentation Roadmap



- First, we'll establish the value that skill positions have, which addresses counter examples.
- Then we'll quantify physical cost, health consequences, and financial vulnerability.
- Finally, we'll prove the feasibility of our arguments and make recommendations

Introduction



- ACL reconstruction hits skill positions harder than QBs: after ACL reconstruction, RBs lose about 20 rushing yards/game and 1.8 yards/attempt, WRs lose about 1.9 yards/target, while QBs show no significant change in performance metrics
- Ankle injuries have multi-year performance effects for RB/WR: a Columbia University Medical Center study found RBs and WRs show a significant decrease in team contribution multiple years after injury, even though they can return to practice.
- According to OverTheCap, QB (Top 10 APY avg) is \$54.36M/year, WR (Top 10 APY avg) is \$32.58M/year, RB (Top 10 APY avg) is \$13.52M/year, and TE (Top 10 APY avg) is \$6.55M/year

EPA Calculations



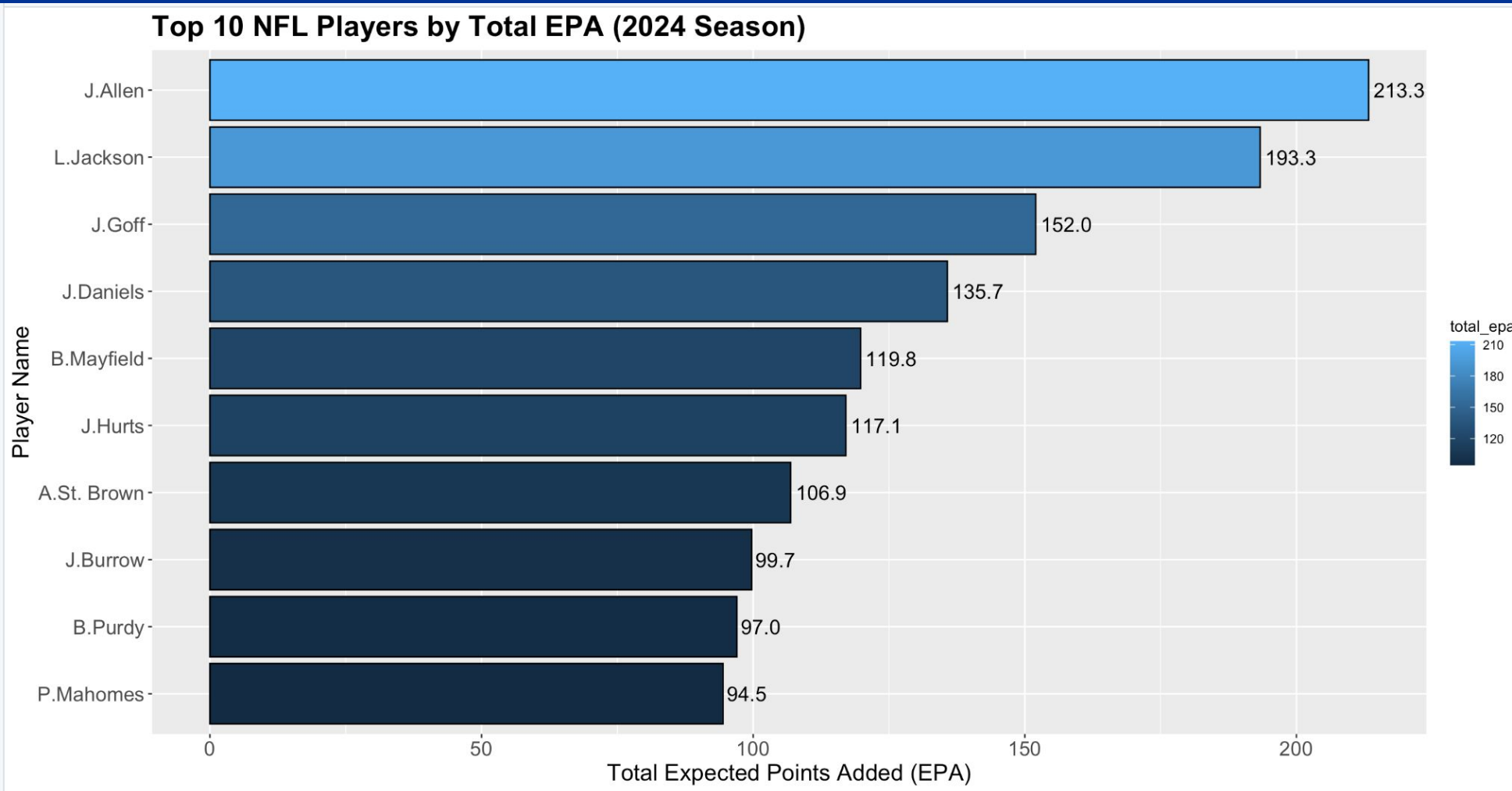
- EPA: Expected points added- illustrates how much a play affects a team's scoring chances
- Use total EPA and EPA/play allows us to compare and quantify the effects that skill position players had on an offense in comparison to quarterbacks.
- Just evaluating players by EPA is a shortcoming, doesn't assign credit to each player
 - Coded an EPA decomposition to try to account for this

Data Cleaning and Filtering

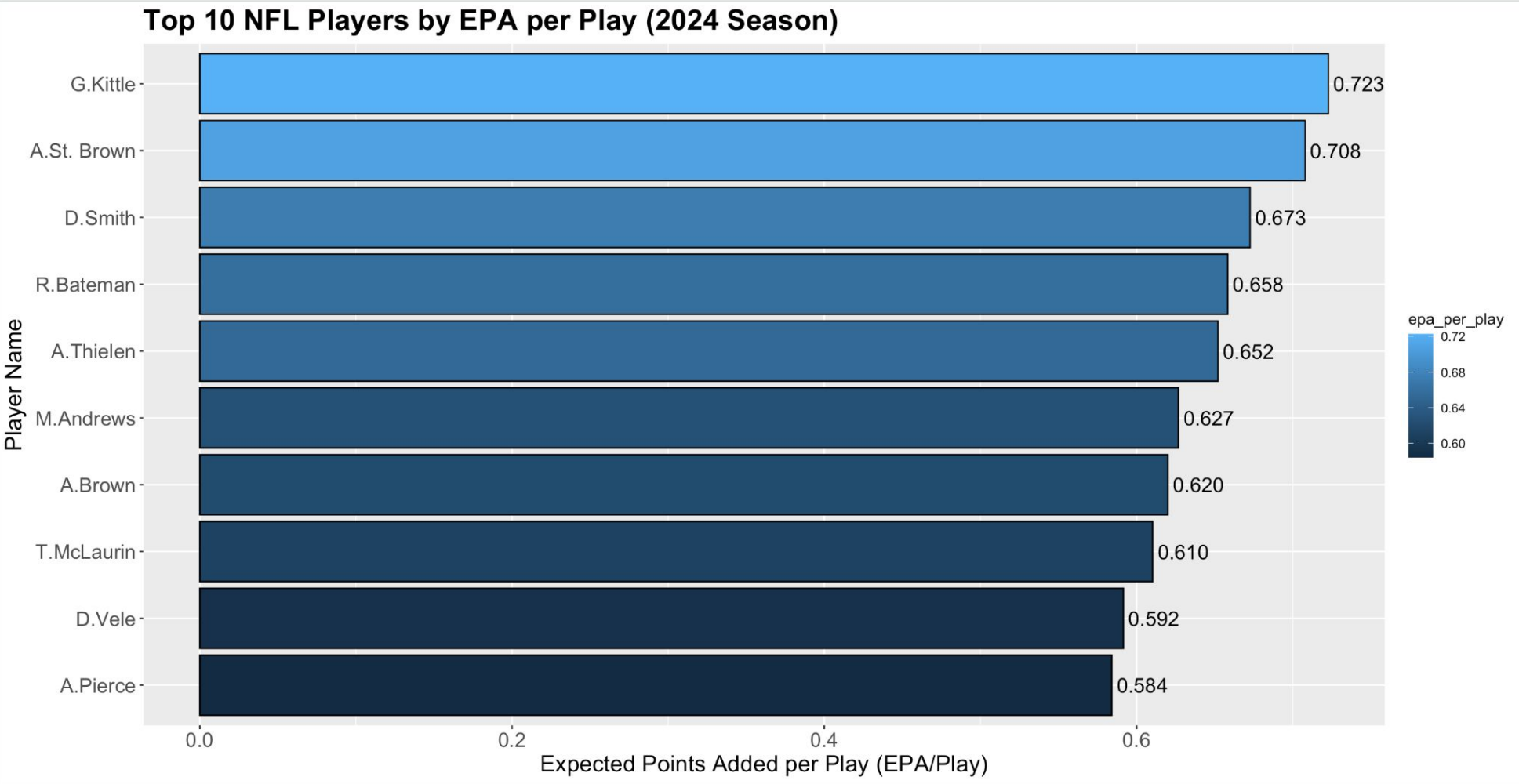


- nflreadr package, includes play by play data from 2024 NFL season
- EPA/play is included in the dataset, but not aggregated by player
- Total EPA for each player encompasses passing EPA, rushing EPA, and receiving EPA
- Filtered to ≥ 50 plays for relevance- must have at least 50 pass attempts, carries, receptions

Total EPA Results



EPA/Play Results



Interpretation of Results



- Total EPA dominated by quarterbacks, EPA/play shows mostly skill players
- On an EPA/play basis, skill position players are extremely efficient
 - Quarterback plays a huge role in this, they also have many more plays/snaps
- Quarterbacks are still objectively more valuable than skill position players, but we argue the gap is not as big as the difference in contract money

EPA Decomposition on Passing Plays - Methods



- Exclusion Criteria: QBs had to have ≥ 350 dropbacks and skill positions had to have ≥ 100 targets
- On all plays, QBs were credited with 100% of the Air EPA
- For YAC EPA, QBs received 25% of the credit on short plays (≤ 5 air yards) and none on long plays
- QBs and WRs were then filtered to only look at players in the top 25% in terms of EPA

EPA Decomposition on Passing Plays - Results



- QBs averaged 0.488 EPA per dropback while skill positions averaged 0.404 EPA with no statistically significant difference at the 95% confidence level (Welch's t-test, $p = 0.0714$)
- While these results do not indicate that skill positions have a greater impact on offense than quarterbacks, they do suggest that the efficiency gap per opportunity between the two may be smaller than their contract values imply.

Player Replaceability



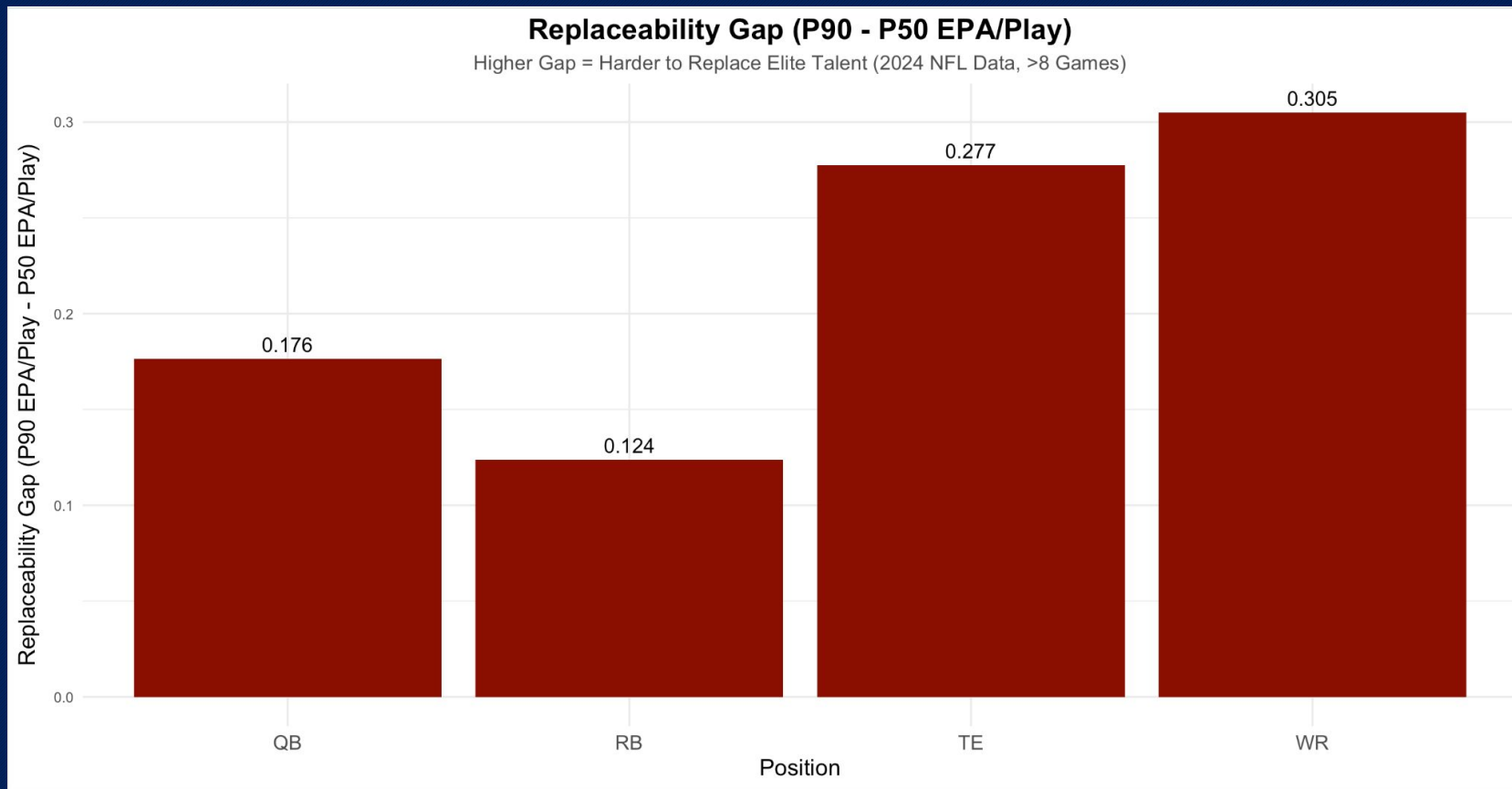
- Goal is to evaluate how difficult it is to replace an elite level player
- Use our previous EPA/play calculations and compare a 90th percentile player(elite) to a 50th percentile player(average)
- Difference in EPA/play represents the difficulty of replacing an elite player(bigger difference = more difficult)

Methodology



- Only looking at NFL starters
 - > 8 games played(at least 50% of the NFL season), over 200 throw attempts for QBs, 90 opportunities for RBs, and 50 opportunities for WRs
- Take the 50th percentile EPA/play and 90th percentile EPA/play for each position using previously calculated EPA data
- Replaceability gap = 90th percentile EPA/play - 50th percentile EPA/play

Player Replaceability Results



- Findings suggest elite TEs and WRs are extremely hard to replace
- EPA does not encompass everything, but we do believe that elite WRs and TEs should be compensated similarly to elite QBs based on their replaceability.
- QBs are still the most valuable position, but their difference in value with skill players is not indicative of the gap in contract value

Contact Score and injury burden - Framing



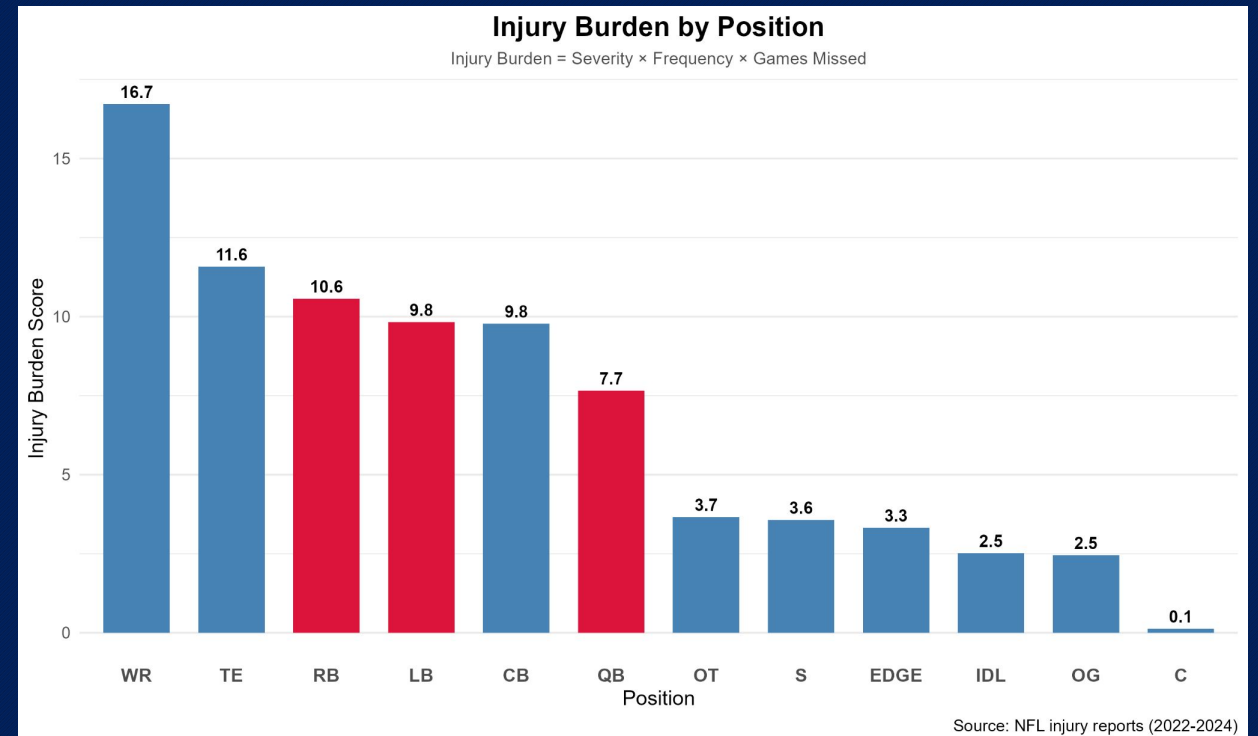
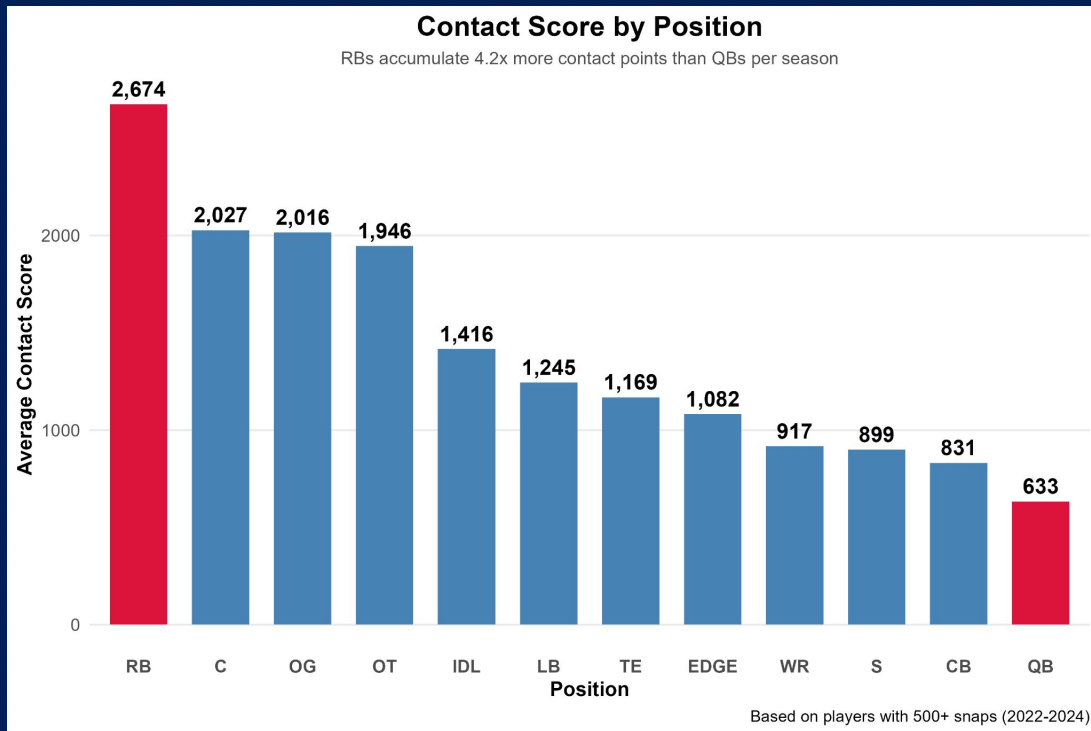
- We wanted to quantify the workload position groups face on average through a season
 - 1. Contact metrics and 2. The injury reports
- The goal was to connect physical demands to compensation
- Data: nflreadr package and NFL Official Injury Reports
- Data cleaning:
 - Name matching across different datasets
 - Position Standardization
 - Snap Filtering

Contact Score and injury burden - Methodology



- $\text{Contact Score} = (\text{Snaps} \times \text{Position Weight}) + (\text{Carries} \times 3) + (\text{Targets} \times 1.5)$
- $\text{Injury Burden} = (\text{Practice Severity}) \times (\text{Injury Frequency}) \times (\text{Games Missed per Player})$

Contact Score and injury burden - Results



Contract Vulnerability - Framing



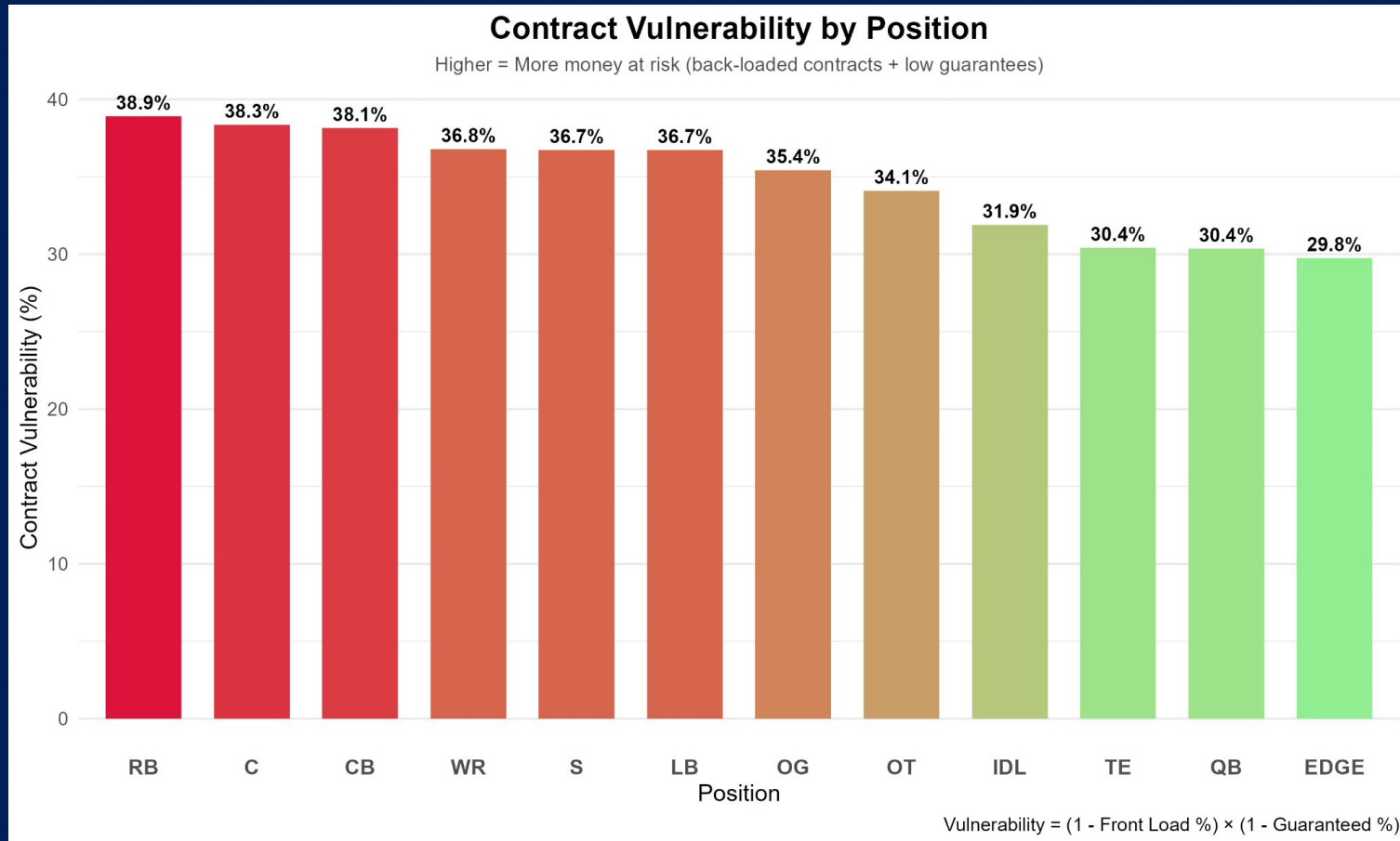
- High contact + high injury $\neq$ exploitation if compensation is adequate
- Need to assess financial protection structures, not just total money
- Question: When injuries happen (and they do for high-contact positions), are players protected?
- Data: Over The Cap database
- Data cleaning:
 - Contract Type Separation
 - Money String Conversion
 - Front-Load Calculation

Contract Vulnerability - Methodology



- Contract Vulnerability = $(1 - \text{Front Load } \%) \times (1 - \text{Guaranteed } \%)$
- Captures Two Risk Dimensions:
 - Back-loading risk $(1 - \text{Front Load } \%)$: Money earned in later years, lost if cut early
 - Guarantee risk $(1 - \text{Guaranteed } \%)$: Non-guaranteed money disappears when injured
- Multiplicative structure: Both factors must be LOW for protection - if either is high, vulnerability increases
- Front Load % = $(\text{Year 0} + \text{Year 1 Salary}) / \text{Total Contract Value}$
- Guaranteed % = $\text{Total Guaranteed Money} / \text{Total Contract Value}$

Contract Vulnerability - Results



Exploitation Score - Framing



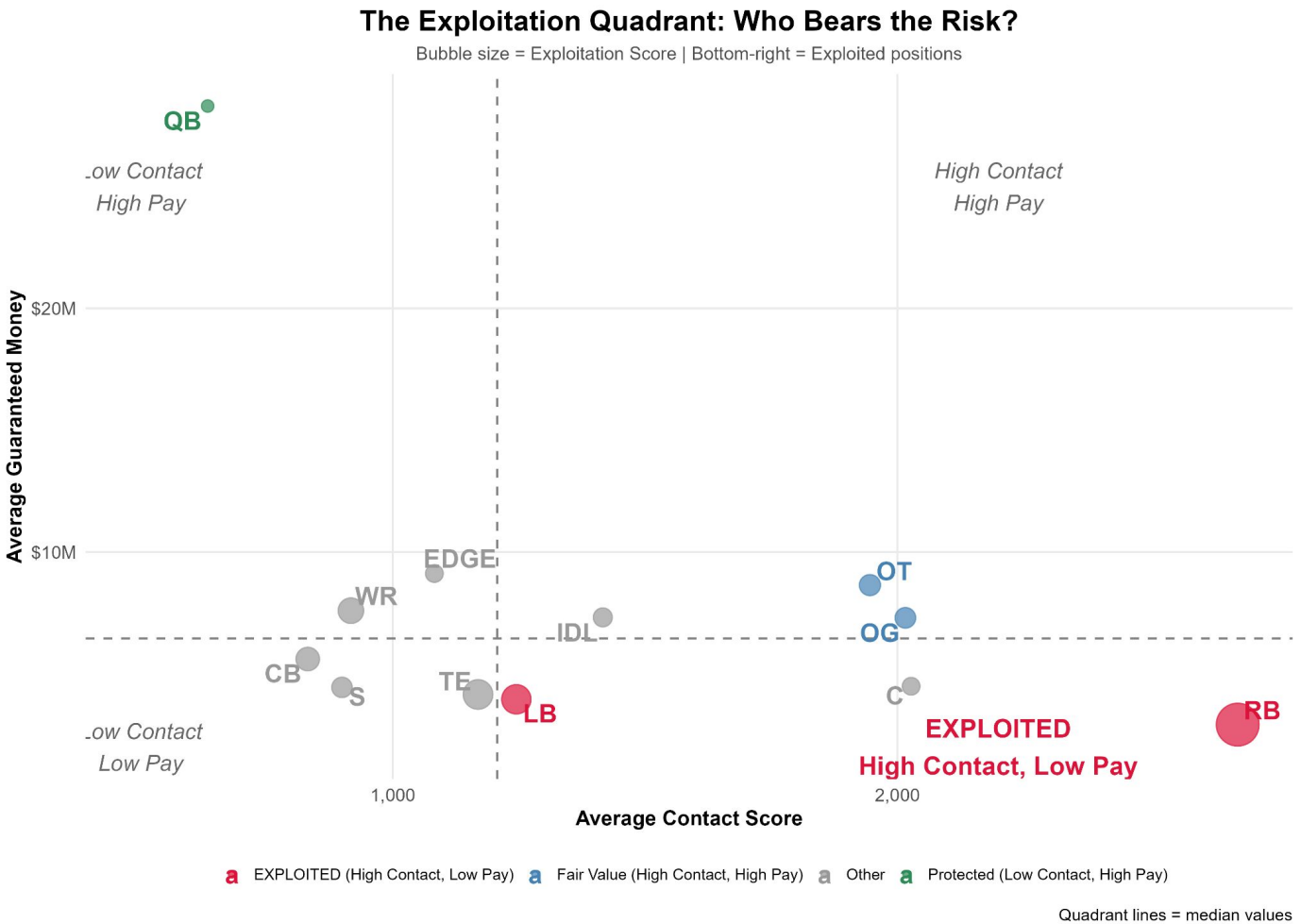
- Need a unified metric connecting all three dimensions.
- Questions to answer:
 - How much physical risk do players bear per dollar of financial protection?
 - Does compensation structure reflect the cumulative workload burden?
 - Are high-contact positions systematically under-protected?
- Data and Data cleaning already done for this

Exploitation Score - Methodology



- $\text{Exploitation Score} = (\text{Contact Score} / 1000) \times (\text{Injury Burden} + 1) / (\text{Guaranteed Money} / \$1\text{M})$
- Units of physical/health risk per million dollars of protection
- Higher score = more exploitation (bearing more risk for less protection)
- Interpretation: RB score of 10.62 means they bear 10.62 units of risk per \$1M guaranteed

Exploitation Score - Results



Overall results



The Numbers Tell the Story

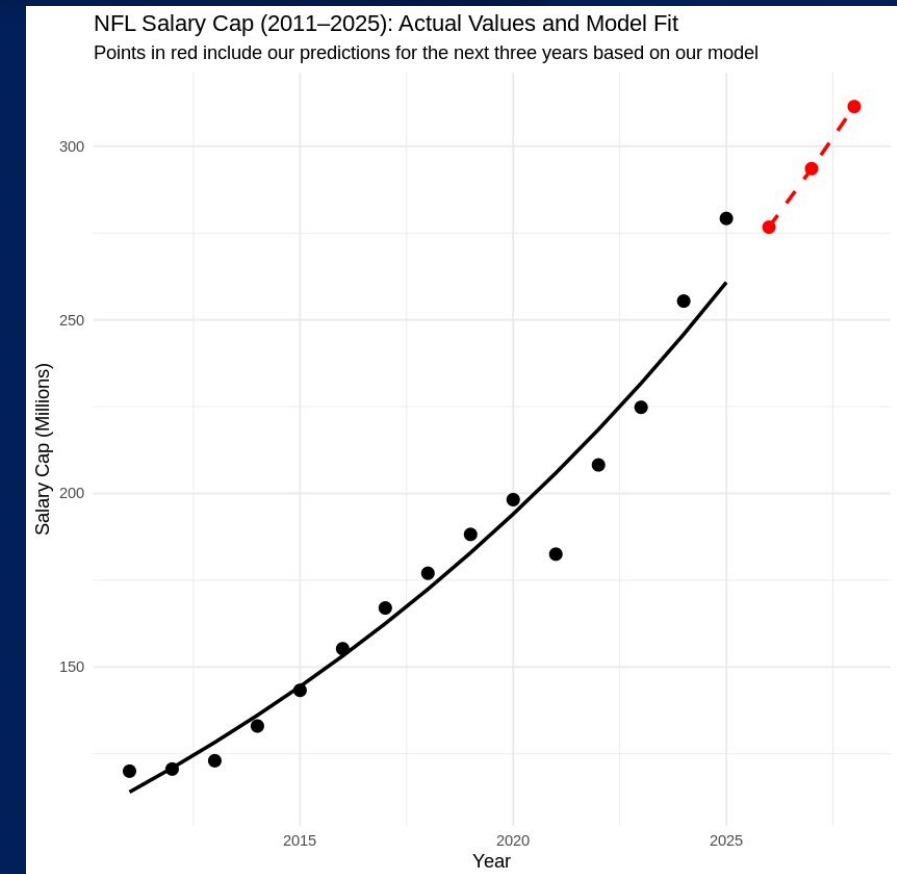
Running backs are systematically exploited: high contact, high injury, low pay

QB Guaranteed Money \$28.3M	QB Contact Score 633	QB Exploitation 0.19
RB Guaranteed Money \$2.9M	RB Contact Score 2,674	RB Exploitation 10.6
Pay Gap 9.7x	Contact Gap 4.2x	Exploitation Gap 55x

Proof of Feasibility - Cap Space



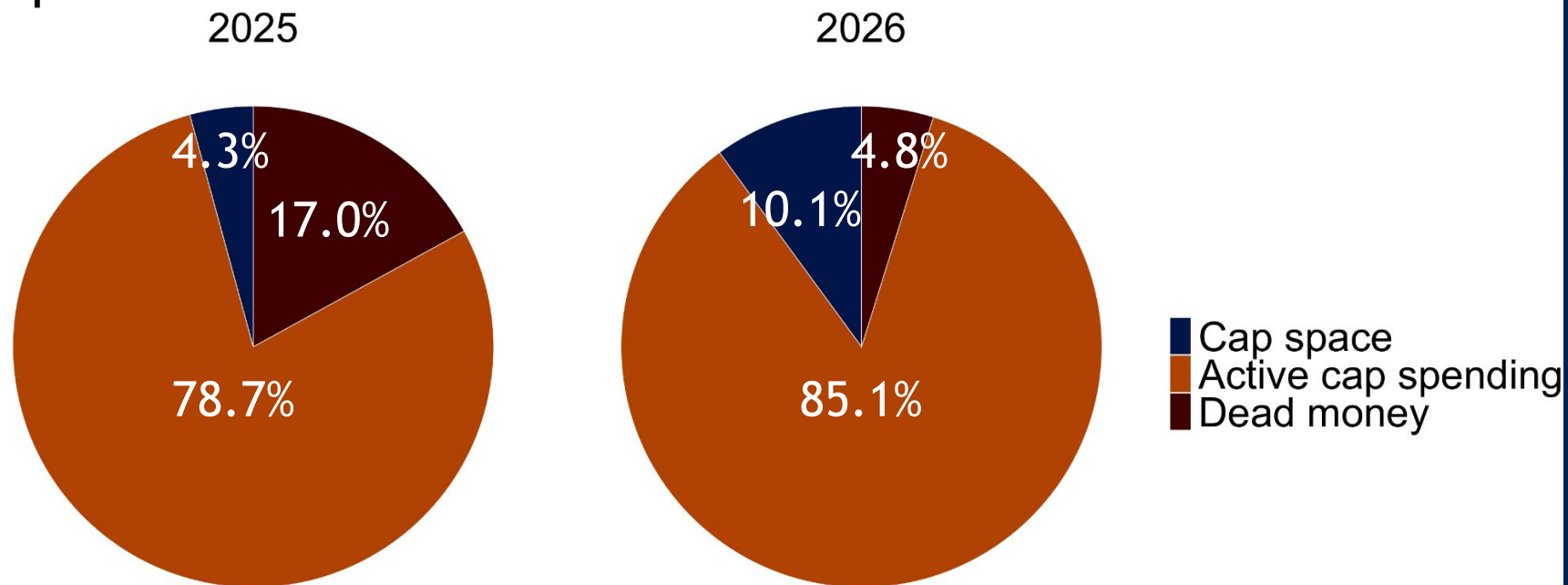
- Using a logarithmic linear regression model, we found the salary cap is expected to increase by roughly 6% every year



Proof of Feasibility - Cap Allocation



Cap Allocation Breakdown



Final Recommendations



- CBA Article 28 (Performance Based Pay)
 - Mainly focuses on Playtime Percentage(player's total plays on offense/defense/special teams and dividing that number by the total plays of the player on the team's roster with the most combined plays on offense, defense, and special teams for that Club.)
 - **Include contact score as a means to calculate performance based pay**
 - Will help RBs in particular to secure more performance based money
- CBA Article 45 Section 3(Injury protection benefits)
 - Currently, players w/ a season ending injury can get paid up to 100% of a player's base salary(maxed out at \$2,100,000 per year)
 - **Amend this with contact score-** WRs, RBs, TEs, LBs making > 2.1M/year should be compensated more
 - 2022 study from Burgess shows skill positions have significant reduction in production compared to quarterbacks following ACL reconstruction surgery

Limitations and Future Work



- In terms of EPA Calculations, it is difficult to assign credit with certainty due to the impact of offensive lineman on offensive production as well as non-targeted skill positions who might be blocking on a given play. Tight ends are especially important in terms of blocking which is not accounted for in our calculations.
- Using nflWAR package or a Markov Chain would help improve precision of results- idea is to be able to assign credit to specific players using EPA